Execution evidence: ingestion, validation, preparation, and feature management



Parquet features | interaction-count popularity + truncated SVD | ranking and rating metrics

1. Data ingestion

data/raw/ml-latest-small/*.csv

```
Ratings source: 100,000 interactions
Movies: 7,312 | Links: 7,312
Columns loaded: userId, movieId, rating, timestamp
Source preview read from the ingested CSV
```

2. Validation and quarantine

outputs/validation_report.json

```
Status: passed
Valid: 100,000
Quarantined: 0
Referential integrity: {'movies': True, 'links': True}
```

3. Preparation

data/processed/train.csv, test.csv, user_item_matrix.npz

```
Train rows: 80,000 | Test rows: 20,000
Chronological: True
Last train timestamp: 1646884202
First test timestamp: 1646884248
```

4. Feature management

data/features/*.parquet and docs/feature_catalog.csv

```
users: 23,188 rows, 7 columns
items: 7,312 rows, 6 columns
interactions: 80,000 rows, 6 columns
Parquet snapshots read from the feature store
```

Visual evidence compiled from the successful run's generated files and metrics.

Execution evidence: model training, evaluation, tracking, and orchestration

5. Model training

models/*.pkl

```
popularity.pkl: present
truncated_svd.pkl: present
Artifacts checked on disk after model training
```

6. Evaluation

outputs/model_comparison.csv

```
popularity: RMSE 1.0134, MAE 0.7617, P@10 0.0273, R@10 0.2270, NDCG@10
0.1181
truncated_svd: RMSE 1.0262, MAE 0.7848, P@10 0.0038, R@10 0.0279, NDCG@10
0.0148
Values read from this run's model comparison
```

7. Experiment tracking and reporting

outputs/experiment_tracking.jsonl and reports/final_report.pdf

```
Runs recorded: 48 total
popularity: run 20261111T131045.302902Z
truncated_svd: run 20261111T131050.657601Z
Model metrics and artifact paths are recorded per run
This PDF includes the stage evidence pages
```

8. Orchestration

src/scheduler.py and dags/recommendation_pipeline.py

```
Pipeline status: passed
Python scheduler: one-shot or repeating interval
Interval option: --interval-hours
Airflow schedule: @daily
DAG configuration evidence (not an Airflow execution log)
```